

JEE MAINS QFT-21 2025



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Or do the failure haunt you....

I know the failure haunts you.... ✨



methongo

By: C I P H E R

Question Paper

Physics Single Correct (Maximum Marks: 80)

Question No. 1 (Only One Option Correct Type)

A faulty thermometer has ice point at -5°C and steam point at 105°C . What would be the temperature shown by this thermometer of human body? (Assume correct temperature of human body is 37°C)

- 34.8°C
- 39.3°C
- 35.7°C
- 41.5°C

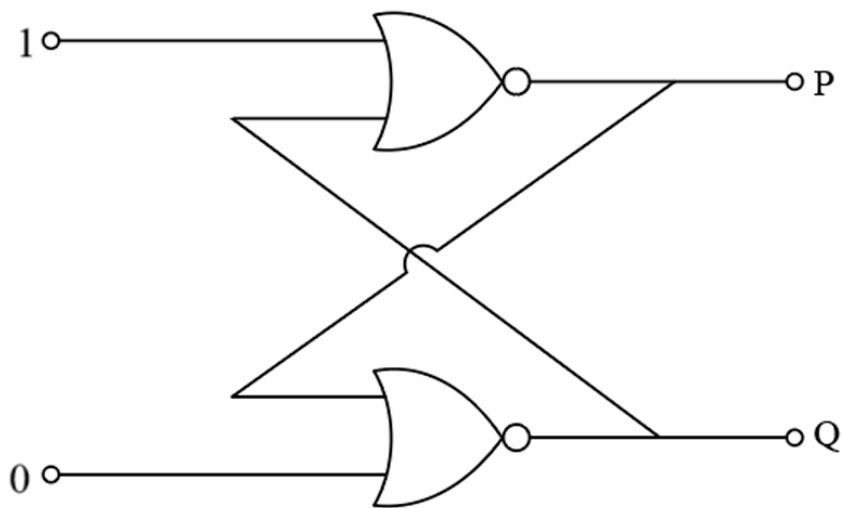
Question No. 2 (Only One Option Correct Type)

Two seconds after projection, a projectile is traveling in a direction inclined at 30° to the horizontal. After one more sec, it is traveling horizontally, the magnitude and direction of its velocity are

- $2\sqrt{2} \text{ m s}^{-1}, 60^{\circ}$
- $20\sqrt{3} \text{ m s}^{-1}, 60^{\circ}$
- $6\sqrt{40} \text{ m s}^{-1}, 30^{\circ}$
- $40\sqrt{6} \text{ m s}^{-1}, 30^{\circ}$

Question No. 3 (Only One Option Correct Type)

In the following circuit, what are P and Q ?



- $P = 0, Q = 1$
- $P = 0, Q = 0$
- $P = 1, Q = 1$
- $P = 1, Q = 0$

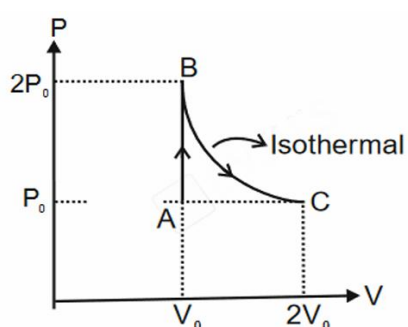
Question No. 4 (Only One Option Correct Type)

In Young's Double slit experiment, a thin glass mica strip of thickness ($t = 8\lambda$) is pasted in front of slit S_1 . If the same strip is now shifted in front of other slit S_2 . If the number of fringes which will cross the central point on the screen is N . Find N ? ($d \gg D$, $\lambda \ll d$ and $n_{\text{mica}} = 1.5$)

- 3
- 6
- 8
- 4

Question No. 5 (Only One Option Correct Type)

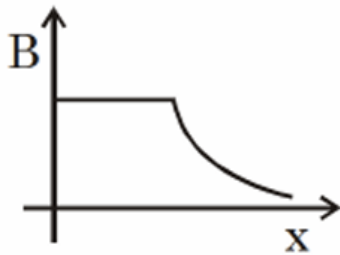
One mole of a diatomic ideal gas undergoes a process shown in $P - V$ diagram. The total heat given to the gas ($\ln 2 = 0.7$) is



- $2.5P_0 V_0$
- $3.9P_0 V_0$
- $1.1P_0 V_0$
- $1.4P_0 V_0$

Question No. 6 (Only One Option Correct Type)

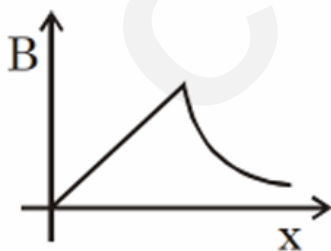
An infinite cylindrical current carrying wire carries a current that is uniformly distributed over its cross section. The magnetic field as a function of distance from it's axis is :-



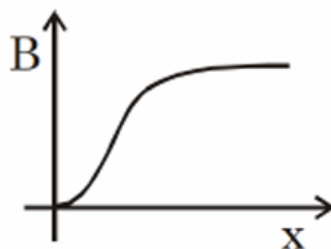
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Question No. 7 (Only One Option Correct Type)

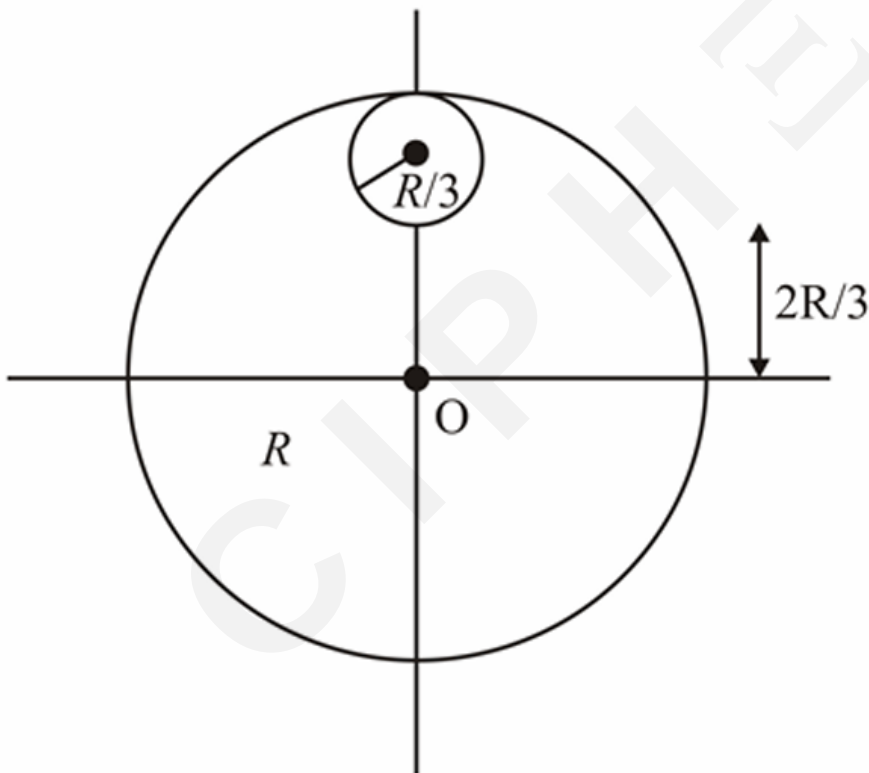
Ultraviolet light of wavelength 99 nm falls on a metal plate of work function 1.0 eV . If

the mass of the electron is 9.1×10^{-31} kg, the wavelength of the fastest photoelectron emitted is

- 0.63 nm
- 0.66 nm
- 0.33 nm
- 0.36 nm

Question No. 8 (Only One Option Correct Type)

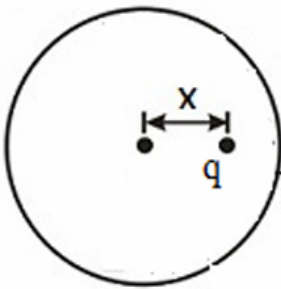
From a circular disc of radius R and mass $9M$, a small disc of radius $R/3$ is removed from the disc. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through O is



- $4MR^2$
- $\frac{40}{9}MR^2$
- $10MR^2$
- $\frac{37}{9}MR^2$

Question No. 9 (Only One Option Correct Type)

A point charge q is kept inside an uncharged conducting spherical shell. The radius of the shell is R and the charge is kept inside the shell at a distance x from the centre. Identify the correct statement.



- Net force acting on the shell is zero
- Net force on the shell is towards right side
- Net force on the shell is towards left side
- Nothing can be said about the net direction of the net force on the shell.

Question No. 10 (Only One Option Correct Type)

Statement - 1: In an isolated conductor, free electrons keep on moving, but no net magnetic force acts on a conductor in a magnetic field.

Statement - 2: In a conductor, the average velocity of thermal motion of electrons is zero. Hence, no current flows through the conductor.

- Statement-1 is True, statement-2 is True: Statement-2 is a correct explanation for statement-1.
- Statement-1 is True, Statement-2 is True: Statement-2 is not a correct explanation for Statement-1.
- Statement-1 is True. Statement-2 is False.
- Statement-1 is False,, Statement-2 is True.

Question No. 11 (Only One Option Correct Type)

A body is executing simple harmonic motion. At a displacement x from mean position, its potential energy is $E_1 = 2J$ and at a displacement y from mean position, its potential energy is $E_2 = 8J$. The potential energy E at a displacement $(x + y)$ from mean position is

- 10 J
- 14 J
- 18 J
- 4 J

Question No. 12 (Only One Option Correct Type)

Remote sensing satellites move in an orbit that is at an average height of about 500 km from the surface of the earth. The camera onboard one such satellite has a screen of area A on which the images captured by it are formed. If the focal length of the camera lens is 50 cm, then the terrestrial area that can be observed from the satellite is close to

- $2 \times 10^3 A$
- $10^6 A$
- $10^{12} A$
- $4 \times 10^{12} A$

Question No. 13 (Only One Option Correct Type)

Within a spherical charge distribution of charge density $\rho(r)$, N equipotential surfaces of potential $V_0, V_0 + \Delta V, V_0 + 2\Delta V, \dots, V_0 + N\Delta V$ ($\Delta V > 0$), are drawn and have increasing radii $r_0, r_1, r_2, \dots, r_N$, respectively. If the difference in the radii of the surfaces is constant for all values of V_0 and ΔV then :

- $\rho(r) = \text{constant}$
- $\rho(r) \propto \frac{1}{r^2}$
- $\rho(r) \propto \frac{1}{r}$
- $\rho(r) \propto r$

Question No. 14 (Only One Option Correct Type)

The correct ascending order of the wavelengths of the following electromagnetic waves is Ultraviolet, X-rays, infrared, gamma rays and visible light.

- X-ray, gamma ray, ultraviolet, infrared, visible light.
- Gamma ray, X-ray, ultraviolet, visible light, infrared.
- Gamma ray, X-ray, infrared, visible light, ultraviolet.
- Gamma ray, X-ray, ultraviolet, infrared, visible light.

Question No. 15 (Only One Option Correct Type)

A gas bubble from an explosion under water oscillates with a time period T , depends upon static pressure p , density of water ρ and the total energy of explosion E . Find the expression for the time period T . (where, k is a dimensionless constant)

- $T = kp^{-5/6} \rho^{1/2} E^{1/3}$
- $T = kp^{-4/7} \rho^{1/2} E^{1/3}$
- $T = kp^{-5/6} \rho^{1/2} E^{1/2}$
- $T = kp^{-4/7} \rho^{1/3} E^{1/2}$

Question No. 16 (Only One Option Correct Type)

A bowl filled with very hot soup cools from 98°C to 86°C in 2 minutes when the room temperature is 22°C . How long it will take to cool from 75°C to 69°C ?

- 2 minutes
- 1.4 minutes
- 0.5 minute
- 1 minute

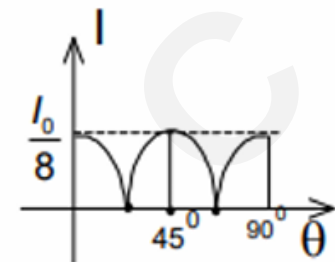
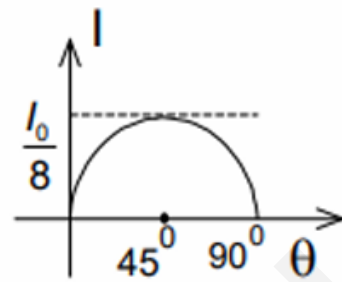
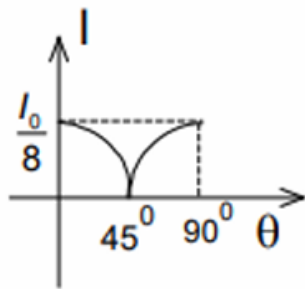
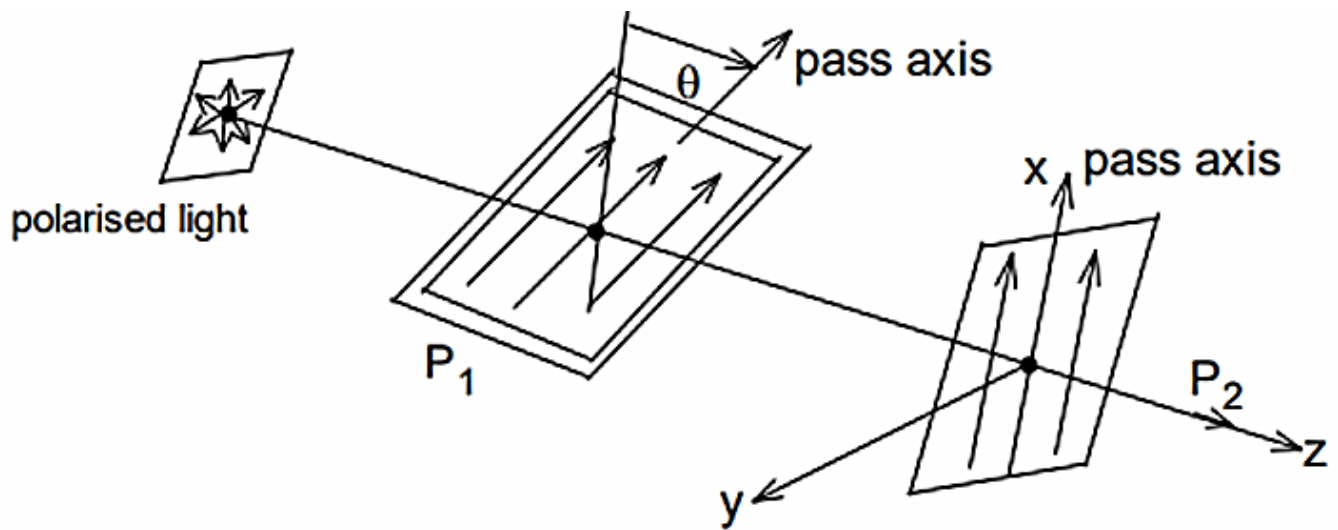
Question No. 17 (Only One Option Correct Type)

A solid sphere rolls down without slipping on a rough inclined plane of inclination $\sin^{-1}(0.42)$. If the acceleration due to gravity is 10 ms^{-2} , the acceleration of the rolling sphere is

- 4.5
- 3
- 9
- 6

Question No. 18 (Only One Option Correct Type)

If a polarised light of intensity $I_0/2$ is passed through two polaroids P_1 and P_2 as shown in figure. Then the intensity of light observed by an observer on other side of P_2 (if the polaroid P_1 is rotated clockwise) as a function of angle θ



- none of these

Question No. 19 (Only One Option Correct Type)

Assertion : When a bottle of cold carbonated drink is opened, a slight fog forms around the opening.

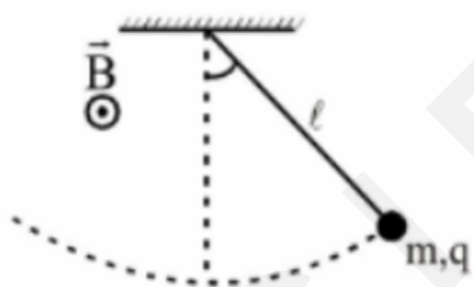
Reason : Adiabatic expansion of the gas causes lowering of temperature and condensation of water vapours.

- if both assertion and reason are true and the reason is the correct explanation of the assertion
- if both assertion and reason are true and the reason is not the correct explanation of the assertion
- if assertion is true but reason is false
- if both assertion and reason are false statements.

Question No. 20 (Only One Option Correct Type)

Statement-1 : A pendulum made of a nonconducting rigid massless rod of length ℓ is attached to a small sphere of mass m and charge q . The pendulum is undergoing oscillations of small amplitude having time period T . Now a uniform horizontal magnetic field $\vec{B}$ out of plane of page is switched on. As a result of this change, the time period of oscillations will change.

Statement-2 : In the situation of statement-1, after the magnetic field is switched on the tension in string will change (except when the bob is at extreme position)



- Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
- Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
- Statement-1 is true, statement-2 is false.
- Statement-1 is false, statement-2 is true.

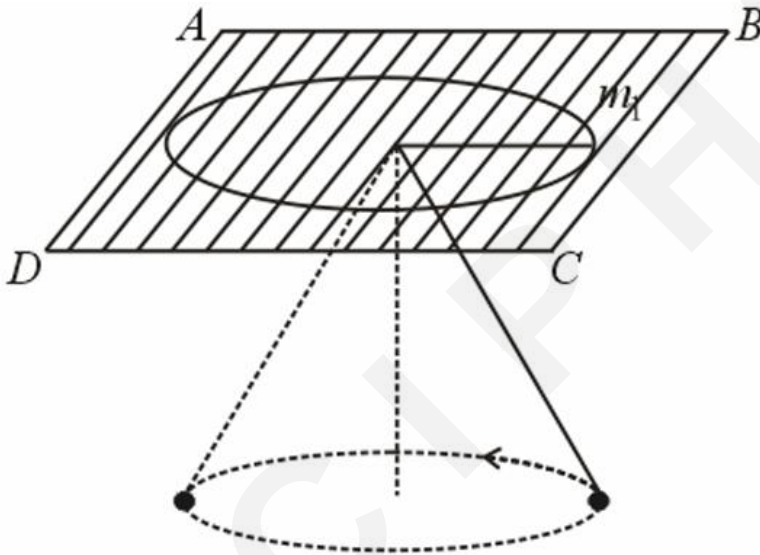
Physics Numerical (Maximum Marks: 20)

Question No. 1 (Numerical Type)

A ray of light is falling on a glass sphere of $\mu = \sqrt{3}$ such that the incident ray and the emergent ray, when produced, intersect at a point on the surface of the sphere. The angle of incidence in degrees is found to be $\frac{2\pi}{k}$. Find the value of k .

Question No. 2 (Numerical Type)

$ABCD$ is a smooth horizontal fixed plane on which mass $m_1 = 0.1 \text{ kg}$ is moving in a circular path of radius 1 m . It is connected by an ideal string which is passing through a smooth hole and connects mass $m_2 = \frac{1}{\sqrt{2}} \text{ kg}$ at the other end as shown. m_2 also moves in a horizontal circle of same radius of 1 m with a speed of $\sqrt{10} \text{ m s}^{-1}$. If $g = 10 \text{ m s}^{-2}$, then the speed of m_1 is—

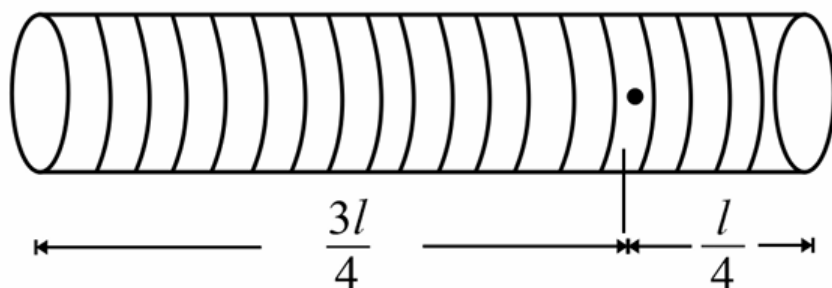


Question No. 3 (Numerical Type)

A solid sphere of radius R made of a material of bulk modulus B is surrounded by a liquid in a cylindrical container. A massless piston of area A (the area of container is also A) floats on the surface of the liquid. When a mass M is placed on the piston to compress the liquid, fractional change in radius of the sphere is $\frac{Mg}{\alpha AB}$. Find the value of α .

Question No. 4 (Numerical Type)

A tightly wound solenoid of radius ' a ' and length ' l ' has n turns per unit length. It carries an electric current i . Magnetic field at a distance $\frac{l}{4}$ from one of the end (inside the solenoid on its axis) is $B = \frac{\mu_0 ni(\sqrt{5}+3)}{\sqrt{K}}$ for $l = 4a$. Find the value of K .



Question No. 5 (Numerical Type)

The radius of the planet is double that of the earth, but their average densities are same. V_p and V_E are the escape velocities of planet and earth respectively. If $\frac{V_p}{V_E} = x$, the value of 'x' is

Chemistry Single Correct (Maximum Marks: 80)

Question No. 1 (Only One Option Correct Type)

The quantum number of four electrons are given below:

- I. $n = 4, l = 2, m_l = -2, m_s = -1/2$
- II. $n = 3, l = 2, m_l = 1, m_s = +1/2$
- III. $n = 4, l = 1, m_l = 0, m_s = +1/2$
- IV. $n = 3, l = 1, m_l = 1, m_s = -1/2$

The correct order of their increasing energies will be:

- $I < III < II < IV$
- $IV < II < III < I$
- $I < II < III < II$
- $IV < III < II < I$

Question No. 2 (Only One Option Correct Type)

	Column-I Complex		Column-II Structure / Geometry / Property
(A)	$[\text{Ni}(\text{CN})_4]^{2-}$	(I)	Tetrahedral / Paramagnetic
(B)	$[\text{Ni}(\text{CO})_4]$	(II)	Tetrahedral / Diamagnetic
(C)	$[\text{NiCl}_4]^{2-}$	(III)	Square planar / Diamagnetic

<i>A</i>	<i>B</i>	<i>C</i>
<i>II</i>	<i>I</i>	<i>III</i>

<i>A</i>	<i>B</i>	<i>C</i>
<i>I</i>	<i>II</i>	<i>III</i>

<i>A</i>	<i>B</i>	<i>C</i>
<i>III</i>	<i>II</i>	<i>I</i>

<i>A</i>	<i>B</i>	<i>C</i>
<i>III</i>	<i>I</i>	<i>II</i>

Question No. 3 (Only One Option Correct Type)

The dipole moment of AX_3 , BX_3 and CX_3 are 1.5 D, 0.5 D and 0 D respectively. The possible shapes of molecules may be (consider C has no Lone pair) (A, B and C are more electronegative than X)

- Pyramidal, T-shape, Trigonal planar respectively.
- T-shape, Pyramidal, Square planar respectively.
- T-shape, Pyramidal, Trigonal planar respectively.
- Pyramidal, T-shape, Square planar respectively.

Question No. 4 (Only One Option Correct Type)

A solution containing a group-IV cation gives a precipitate on passing H_2S . A solution of this precipitate in dil. HCl produces a white precipitate with NaOH solution and bluish-white precipitate with basic potassium ferrocyanide. The cation is :

- Co^{2+}
- Ni^{2+}
- Mn^{2+}
- Zn^{2+}

Question No. 5 (Only One Option Correct Type)

Given two statements below :

Statement I : In Cl_2 molecule the covalent radius is double of the atomic radius of chlorine.

Statement II : Radius of anionic species is always greater than their parent atomic radius.

Choose the most appropriate answer from options given below

- Both Statement I and Statement II are correct.
- Both Statement I and Statement II are incorrect.
- Statement I is correct but Statement II is incorrect.
- Statement I is incorrect but Statement II is correct.

Question No. 6 (Only One Option Correct Type)

List – I		List – II	
(A)	Tartaric acid	(P)	Metamerism
(B)	$\text{CH}_3\text{CH}=\text{CH}-\text{CH}_3$	(Q)	Cis-trans isomerism
(C)	$\text{CH}_3-\text{CH}_2-\text{NO}_2$	(R)	Optical isomerism
(D)	$\text{CH}_3-\text{CH}_2-\text{O}-\text{CH}_2-\text{CH}_3$	(S)	Conformational isomerism
		(T)	Dynamic isomerism

- (A – R)(B – Q) (C – T)(D – P)
- (A – P)(B – Q) (C – T) (D – S)
- (A – R)(B – T)(C – Q)(D – P)
- (A – R)(B – S)(C – T)(D – P)

Question No. 7 (Only One Option Correct Type)

Given $E^\circ_{\text{Au}^{3+}/\text{Au}} = 1.52 \text{ V}$ and $E^\circ_{\text{Au}^{3+}/\text{Au}^+} = 1.36 \text{ V}$

Point out the correct statement of the following

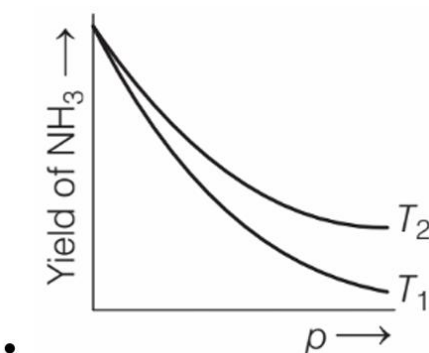
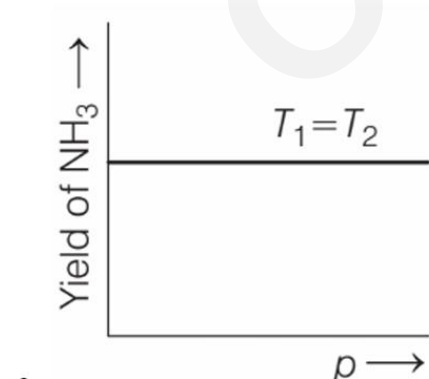
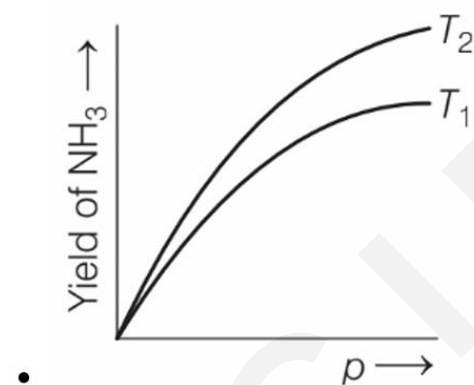
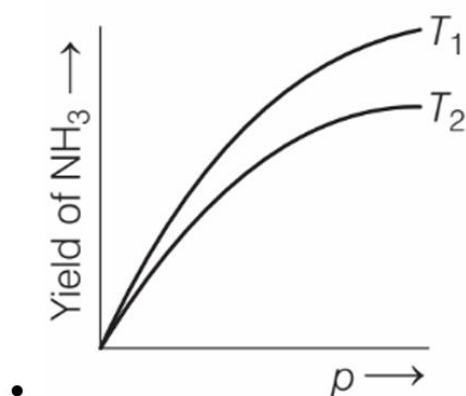
- Au^{3+} disproportionates into Au^{4+} and Au^{2+} in aqueous solution
- Au^{3+} disproportionates into Au^{4+} and Au^+ in aqueous solution

- Au^+ disproportionates into Au^{3+} and Au in aqueous solution
- Au^+ disproportionates into Au^{2+} and Au in aqueous solution

Question No. 8 (Only One Option Correct Type)

For the formation of $\text{NH}_3(g)$ from H_2 and N_2 which one of the following graphs represents the yield of ammonia at equilibrium?

(Assume $T_1 < T_2$, unless mentioned)



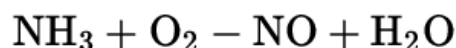
Question No. 9 (Only One Option Correct Type)

When H_2S is passed through a solution of CoCl_2 , a black precipitate is formed. It is soluble in aqua regia. On adding acetic acid in excess and potassium nitrate, yellow precipitate is formed. The oxidation number of cobalt in this compound is $+n$. Find the value of n .

- +1
- +2
- +4
- +3

Question No. 10 (Only One Option Correct Type)

The given reaction is an important step in Ostwald's method for manufacturing of HNO_3 . If we start with 6.8 g of NH_3 and 40 g of O_2 , then what mass % of excess reagent will be left behind ?



- 45%
- 50%
- 60%
- 70%

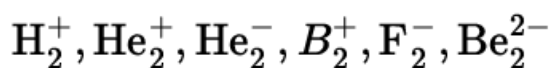
Question No. 11 (Only One Option Correct Type)

Arrange the following cyano complexes in decreasing order of their magnetic moment.

- $[\text{Cr}(\text{CN})_6]^{3-} > [\text{Mn}(\text{CN})_6]^{3-} > [\text{Fe}(\text{CN})_6]^{3-} > [\text{Co}(\text{CN})_6]^{3-}$
- $[\text{Mn}(\text{CN})_6]^{3-} > [\text{Cr}(\text{CN})_6]^{3-} > [\text{Fe}(\text{CN})_6]^{3-} > [\text{Co}(\text{CN})_6]^{3-}$
- $[\text{Fe}(\text{CN})_6]^{3-} > [\text{Cr}(\text{CN})_6]^{3-} > [\text{Mn}(\text{CN})_6]^{3-} > [\text{Co}(\text{CN})_6]^{3-}$
- $[\text{Co}(\text{CN})_6]^{3-} > [\text{Cr}(\text{CN})_6]^{3-} > [\text{Mn}(\text{CN})_6]^{3-} > [\text{Fe}(\text{CN})_6]^{3-}$

Question No. 12 (Only One Option Correct Type)

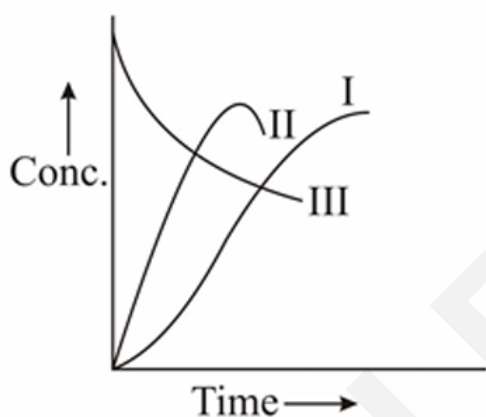
How many among the given species have a bond order of 0.5 ?



- 4
- 5
- 3
- 2

Question No. 13 (Only One Option Correct Type)

In a CSTR, a homogeneous reaction's first-order of the type $\text{X} \rightarrow \text{Y} \rightarrow \text{Z}$ (consecutive reaction) is carried out. Among the following, which curve shows a variation of concentration of X, Y and Z with time?



- I, II, III
- III, II, I
- III, I, II
- II, III, I

Question No. 14 (Only One Option Correct Type)

The correct match between Item-I (starting material) and Item-II (reagent) for the preparation of benzaldehyde is:

Item-I		Item-II
(I)	Benzene	(P) HCl and SnCl_2 , H_3O^+
(II)	Benzonitrile	(Q) H_2 , $\text{Pd} - \text{BaSO}_4$, S and quinoline
(III)	Benzoyl Chloride	(R) Co , HCl and AlCl_3

- ☐ (I)-(Q), (II)-(R) and (III)-(P)
- ☐ (I)-(P), (II)-(Q) and (III)-(R)
- ☐ (I)-(R), (II)-(P) and (III)-(Q)
- ☐ (I)-(R), (II)-(Q) and (III)-(P)

Question No. 15 (Only One Option Correct Type)

The sublimation energy of $\text{I}_2(\text{s})$ is 57.3 kJ mol^{-1} and the enthalpy of fusion is 15.5 kJ mol^{-1} . The enthalpy of vaporization of I_2 is

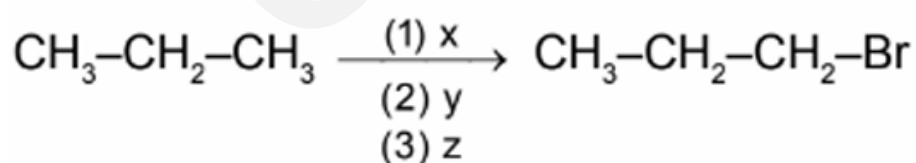
- 41.8 kJ mol^{-1}
- $-41.8 \text{ kJ mol}^{-1}$
- 72.8 kJ mol^{-1}
- $-72.8 \text{ kJ mol}^{-1}$

Question No. 16 (Only One Option Correct Type)

A trend which is common to elements of both the group 1^{st} and group 17^{th} , on going from top to bottom

- Boiling point increases
- Electorn affinity increases
- Oxidising power increases
- Ionization energy decreases.

Question No. 17 (Only One Option Correct Type)



x, y, z are respectively :

- $\text{Br}_2/h\nu$, conc. H_2SO_4 , ThO_2/Δ
- $\text{Cl}_2/h\nu$, alc. KOH/Δ , $\text{HBr}/\text{R}_2\text{O}_2$
- $\text{Cl}_2/h\nu$, $\text{tBuO}^\ominus/\Delta$, HBr (dark)
- $\text{F}_2/h\nu$, $\text{Al}_2\text{O}_3/\Delta$, NBS

Question No. 18 (Only One Option Correct Type)

Amine is not formed in the reaction

I. Hydrolysis of RCN

II. Reduction of $\text{RCH}=\text{NOH}$

III. Hydrolysis of RNC

IV. Hydrolysis of RCONH_2

The correct answer is

- I, II and IV
- I and IV
- II and III
- I, II and III

Question No. 19 (Only One Option Correct Type)

Match List-I with List-II and select the correct answer using the codes given below the lists.

List-I		List-II
(p)	Nucleic acids	1.DNA
(q)	Uracil	2.Hormones
(r)	Thymine	3.Polynucleotides
(s)	Double-helix structure	4.RNA

- ☐ (p)-4, (q)-3, (r)-1, (s)-1
- ☐ (p)-3, (q)-4, (r)-1, (s)-1
- ☐ (p)-3, (q)-1, (r)-4, (s)-1
- ☐ (p)-3, (q)-1, (r)-1, (s)-4

Question No. 20 (Only One Option Correct Type)

When 400 mL of 0.2 M H_2SO_4 solution is mixed with 600 mL of 0.1 M NaOH solution, the increase in temperature of the final solution is $\text{_____} \times 10^{-2} \text{ K}$. (Round off to the nearest integer).

[Use $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O} : \Delta_r H = -57.1 \text{ kJ mol}^{-1}$]

Specific heat of $\text{H}_2\text{O} = 4.18 \text{ J K}^{-1} \text{ g}^{-1}$, density of $\text{H}_2\text{O} = 1.0 \text{ g cm}^{-3}$

Assume no change in volume of solution on mixing.

- 76
- 82
- 28

Chemistry Numerical (Maximum Marks: 20)

Question No. 1 (Numerical Type)

In Duma's method for estimation of nitrogen 0.125 g of an organic compound gave 20 ml of nitrogen collected at 300 K temperature and 725 mm pressure. If the aqueous tension at 300 K is 25 mm, the percentage of nitrogen in the compound is: (nearest integer)

Question No. 2 (Numerical Type)

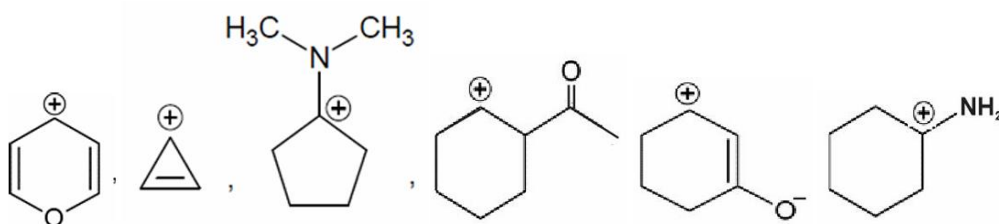
In the saturated aqueous solution of PbCl_2 , the freezing point of water decreases by $\left(\frac{D}{100}\right)^\circ\text{C}$, then 'D' is : (Given k_{sp} of $\text{PbCl}_2 = 4 \times 10^{-6}$, $K_{f, \text{water}} = 2 \text{ K kg/mole}$, assume molality = molarity)

Question No. 3 (Numerical Type)

A first order reaction $\text{A} \rightarrow \text{B}$, requires activation energy of 70 kJ mol^{-1} . When a 20% solution of A was kept at 25°C for 20 minutes, 25% decomposition took place. What will be the percent decomposition in the same time in a 30% solution maintained at 40°C ? Assume that activation energy remains constant in this range of temperature. (Given that $\log 2 = 0.3$, $\log 3 = 0.48$, $\text{antilog } 0.5879 = 3.872$ & $\text{antilog } 0.465 = 2.917$) (nearest integer)

Question No. 4 (Numerical Type)

How many of the following intermediates have at least one contributing structure in which all atoms have their octet complete?



Question No. 5 (Numerical Type)

Apatite is a mineral that is found in tooth enamel. When fluoride tooth pastes are used, this mineral is converted to fluoroapatite, which is more resistant to tooth decay. The formula of apatite is $\text{Ca}_5(\text{PO}_4)_x\text{F}$. What is the value of x , if the compound contains 18.45% by weight of phosphorus? (Atomic weight of $\text{P} = 31$, $\text{Ca} = 40$, $\text{F} = 19$ and $\text{O} = 16$)

Mathematics Single Correct (Maximum Marks: 80)

Question No. 1 (Only One Option Correct Type)

The number of ways of choosing a committee of 2 women & 3 men from 5 women & 6 men, if Mr. A refuses to serve on the committee if Mr. B is a member & Mr. B can only serve, if Miss C is the member of the committee, is

- 124
- 136
- 132
- 128

Question No. 2 (Only One Option Correct Type)

If the system of linear equations

$x + y + z = 5$, $x + 2y + 2z = 6$, $x + 3y + \lambda z = \mu$, ($\lambda, \mu \in \mathbb{R}$), has infinitely many solutions, then the value of $\lambda + \mu$ is:

- 7
- 10
- 12
- 9

Question No. 3 (Only One Option Correct Type)

Let $A = \{(x, y) : x + y = p\}$, $B = \{(x, y) : x^3 + y^3 < p\}$ be two sets such that $A \cap B = \phi$ for any real number p , then the exhaustive intervals of p is

- $[-2, 2]$
- $\{0\} \cup [2, \infty)$
- $(-\infty, -2] \cup \{0\}$
- $\{x! \mid x \in \mathbb{N} \wedge x \leq 3\}$

Question No. 4 (Only One Option Correct Type)

Area bounded by $|x - 3y| + |x + 3y| \leq 12$ and $xy \geq 3$ is

- $4(3 - \ln 2)$
- $8(3 - \ln 2)$
- $12(3 - \ln 2)$
- none of these

Question No. 5 (Only One Option Correct Type)

The midpoint of a chord of the ellipse $x^2 + 4y^2 - 2x + 20y = 0$ is $(2, -4)$. The equation of the chord is

- $x - 6y = 26$
- $x + 6y = 26$
- $6x - y = 26$
- $6x + y = 26$

Question No. 6 (Only One Option Correct Type)

The sum of all the values of $|\lambda|$ such that the lines $x + 2y - 3 = 0$, $3x - y - 1 = 0$ and $2\lambda x + y - 2 = 0$ cannot form a triangle is K, find 100K

- 200
- 225
- 235
- 250

Question No. 7 (Only One Option Correct Type)

In a tetrahedron $OABC$, the edges are of lengths,

$|OA| = |BC| = a$, $|OB| = |AC| = b$, $|OC| = |AB| = c$. Let G_1 and G_2 be the centroids of the triangle ABC and AOC such that $OG_1 \perp BG_2$, then the value of $\frac{a^2+c^2}{b^2}$ is (O being origin)

- 2
- 3
- 6
- 9

Question No. 8 (Only One Option Correct Type)

Let the solution curve of the differential equation $xdy = (\sqrt{x^2 + y^2} + y)dx, x > 0$, intersect the line $x = 1$ at $y = 0$ and the line $x = 2$ at $y = \alpha$. Then the value of α is

- $\frac{1}{2}$
- $\frac{3}{2}$
- $-\frac{3}{2}$
- $\frac{5}{2}$

Question No. 9 (Only One Option Correct Type)

If square root of, $a^{\frac{1}{a}} \cdot (2a)^{\frac{1}{2a}} \cdot (4a)^{\frac{1}{4a}} \cdot (8a)^{\frac{1}{8a}} \dots \infty$ is $\frac{8}{27}$, then the value of 'a' is -

- $\frac{1}{2}$
- $\frac{1}{3}$
- $\frac{1}{4}$
- $\frac{1}{5}$

Question No. 10 (Only One Option Correct Type)

If $\alpha \in \mathbb{R} - \{-1\}$ and $f(x) = |(|x| + \alpha)(|x| - 1)|$, then the number of points at which $f(x)$ is not differentiable, is

- 3 , when $\alpha < 0$
- 5 , when $\alpha > 0$
- 4 , when $\alpha > 0$
- 5 , when $\alpha < 0$

Question No. 11 (Only One Option Correct Type)

The sum of two natural numbers n_1 and n_2 is known to be equal to 100. The probability that their product being greater than 1600 , is equal to

- $\frac{20}{33}$
- $\frac{58}{99}$
- $\frac{13}{33}$
- $\frac{59}{99}$

Question No. 12 (Only One Option Correct Type)

Let two non - collinear vectors $\vec{a}$ and $\vec{b}$ inclined at an angle $\frac{2\pi}{3}$ be such that $|\vec{a}| = 3$ and $|\vec{b}| = 4$. A point P moves so that any time t the position vector $\vec{OP}$ (where O is the origin) is given as $\vec{OP} = (e^t + e^{-t})\vec{a} + (e^t - e^{-t})\vec{b}$. If the least distance of P from origin is $\sqrt{2}\sqrt{\sqrt{a} - b}$ where $a, b \in N$ then find the value of $\frac{(a+b)}{8}$.

- 61
- 65
- 58
- 72

Question No. 13 (Only One Option Correct Type)

If $f(x + y) = f(x) + f(y) - xy - 1, \forall x, y \in R$ and $f(1) = 1$, then the number of solutions of $f(n) = n, n \in N$, is

- one
- two
- no solution
- three

Question No. 14 (Only One Option Correct Type)

Let S_n denotes the sum of first n term of an arithmetic progression whose first term is -4 and common difference is 1 . If $V_n = 2 S_{n+2} - 2 S_{n+1} + S_n (n \in N)$, then

- $V_n = \frac{-9n^2+5n-12}{2}, (n \in N)$
- $V_n = \frac{n^2+5n+12}{2}, (n \in N)$
- Minimum value of V_n is -9
- Minimum value of V_n is $-\frac{73}{8}$

Question No. 15 (Only One Option Correct Type)

Which of the following is an even function?

- $f(x) = \log \left(\frac{1-x}{1+x} \right)$
- $f(x) = \log \left(x + \sqrt{1+x^2} \right)$
- $f(x) = \frac{x}{e^x-1} + \frac{x}{2} + 1$
- $f(x) = e^{2x} + \sin x$

Question No. 16 (Only One Option Correct Type)

The coefficient of t^8 in the expansion of $(1 + 2t^2 - t^3)^9$, is

- 1680
- 2140
- 2520
- 2730

Question No. 17 (Only One Option Correct Type)

Let f, g and h are differentiable function such that $g(x) = f(x) - x$ and $h(x) = f(x) - x^3$ are both strictly increasing functions, then the function $F(x) = f(x) - \frac{\sqrt{3}x^2}{2}$ is

- Strictly increasing $\forall x \in R$
- Strictly decreasing $\forall x \in R$
- Strictly decreasing on $\left(-\infty, \frac{1}{\sqrt{3}}\right)$ and strictly increasing on $\left(\frac{1}{\sqrt{3}}, \infty\right)$
- Strictly increasing on $\left(-\infty, \frac{1}{\sqrt{3}}\right)$ and strictly decreasing on $\left(\frac{1}{\sqrt{3}}, \infty\right)$

Question No. 18 (Only One Option Correct Type)

Given $f(x) = \log_{10} x$ and $g(x) = e^{\pi i x}$, where $i = \sqrt{-1}$. If

$$\phi(x) = \begin{vmatrix} f(x)g(x) & (f(x))^{g(x)} & 1 \\ f(x^2)g(x^2) & (f(x^2))^{g(x^2)} & 0 \\ f(x^3)g(x^3) & (f(x^3))^{g(x^3)} & 1 \end{vmatrix}, \text{ then the value of } \phi(10) \text{ is}$$

- 1
- 2
- 0
- i

Question No. 19 (Only One Option Correct Type)

The length of the latus-rectum of the parabola

$$169 [(x - 1)^2 + (y - 3)^2] = (5x - 12y + 17)^2 \text{ is } K, \text{ find } 13K.$$

- 24
- 28
- 32
- 34

Question No. 20 (Only One Option Correct Type)

The number of real roots of the equation $\sin \left[2 \cos^{-1} \left\{ \cot \left(2 \tan^{-1} x \right) \right\} \right] = 0$ that are greater than or equal to one are

- 1
- 2
- 3
- 4

Mathematics Numerical (Maximum Marks: 20)

Question No. 1 (Numerical Type)

Let $a, b \in \mathbb{R}$. If the mirror image of the point $p(a, 6, 9)$ w.r.t. line $\frac{x-3}{7} = \frac{y-2}{5} = \frac{z-1}{-9}$ is $(20, b, -a - 9)$, then $|a + b|$ is equal to

Question No. 2 (Numerical Type)

If $\int \sqrt{\frac{x-5}{x-7}} dx = A\sqrt{x^2 - 12x + 35} + \log|x - 6 + \sqrt{x^2 - 12x + 35}| + C$ then $A =$

Question No. 3 (Numerical Type)

Consider the differential equation $\left(\frac{2+\sin x}{y+1} \right) \frac{dy}{dx} = -\cos x$. If $y(0) = 1$, then evaluate $6y\left(\frac{\pi}{2}\right) + 5$

Question No. 4 (Numerical Type)

Let $f : A \rightarrow A$ where $A = \{1, 2, 3, 4, 5, 6, 7\}$, then number of functions f such that $f(f(f(x))) = x \forall x \in A$, is:

Question No. 5 (Numerical Type)

Let the circle $(x - 1)^2 + (y - 2)^2 = 25$ cuts rectangular hyperbola with transverse axis along $y = x$ at 4 points A, B, C and D having coordinates $(x_i, y_i), \ell = 1, 2, 3, 4$ such that origin is centre of hyperbola. If

$\ell = x_1 + x_2 + x_3 + x_4, m = x_1^2 + x_2^2 + x_3^2 + x_4^2, n = y_1^2 + y_2^2 + y_3^2 + y_4^2$ then $\frac{m+n}{\ell} =$

(ANSWER KEY)

Mathematics

Q1 []	Q2 []	Q3 []	Q4 []	Q5 []
Q6 []	Q7 []	Q8 []	Q9 []	Q10 []
Q11[]	Q12 []	Q13 []	Q14 []	Q15 []
Q16[]	Q17 []	Q18 []	Q19 []	Q20 []

Q21 []	Q22 []	Q23 []	Q24 []	Q25 []
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Physics

Q1 []	Q2 []	Q3 []	Q4 []	Q5 []
Q6 []	Q7 []	Q8 []	Q9 []	Q10 []
Q11[]	Q12 []	Q13 []	Q14 []	Q15 []
Q16[]	Q17 []	Q18 []	Q19 []	Q20 []

Q21 []	Q22 []	Q23 []	Q24 []	Q25 []
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Chemistry

Q1 []	Q2 []	Q3 []	Q4 []	Q5 []
Q6 []	Q7 []	Q8 []	Q9 []	Q10 []
Q11[]	Q12 []	Q13 []	Q14 []	Q15 []
Q16[]	Q17 []	Q18 []	Q19 []	Q20 []

Q21 []	Q22 []	Q23 []	Q24 []	Q25 []
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Answer Key will appear on its own after you're finished!